Engel’s Law in practice: why richer countries spend *more* on food but *less* of their budget on it

1) Research question and scope

Question. What does the available evidence show about how **food spending levels** and **food spending shares** change with prosperity—and what definitions and measurement choices determine whether comparisons are valid?

Core construct used in the main cross-country evidence. In the Our World in Data (OWID) presentation of USDA ERS data, **“food expenditure” is food purchased for consumption at home only**, explicitly **excluding** out-of-home food (restaurants/cafes), **alcohol**, and **tobacco**, and values are in **current US dollars per person** (not adjusted for inflation or cross-country price levels). These constraints are stated in OWID’s definitions and caveats and bound what the charts can (and cannot) support: [1] (archived copy: [2]).

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2) Executive summary (auditable takeaways)

1. **Cross-country “paradox” resolves cleanly under Engel’s law:** as total consumer expenditure rises, **absolute at-home food spending rises**, while **food’s budget share falls**. OWID frames and visualizes both relationships using USDA ERS data and explicitly notes the restrictive “at-home only” definition and “current US$” caveats ([1]; interactive charts include [3]).
2. **Useful rule-of-thumb thresholds (from OWID’s narrative):**
    - At **total expenditures below \$5,000/person/year**, people **tend to spend < \$1,000/year** on at-home food.
    - At **expenditures above \$10,000/person/year**, **\$2,000+** on at-home food is common.
    - For **shares**, countries below **\$10,000** often spend **≥25%** of total expenditure on (at-home) food; at high expenditure levels, **≤10%** is common—despite higher absolute spending.
      All are stated directly in OWID’s Engel’s law article ([1]).
3. **Within-country evidence points in the same direction:** Engel-type gradients are reported within countries, not only across them—e.g., a multilevel regression analysis across South African districts (Mulamba, 2022: [4]). The report’s China pointer (Chen, 2022) is retained as a lead but is not fully auditable from the materials collected here.

4. **Measurement matters at every step.** Even within one country, different survey instruments can produce systematically different food-at-home spending levels (FoodAPS vs CE vs NHANES), per USDA ERS EIB-157 ([5]). For cross-country work, comparability also depends on **classification boundaries** (e.g., "food at home" vs "restaurants") and on whether comparisons use **nominal current US$** or **PPP/price-level–adjusted** measures (Eurostat–OECD PPP Manual DOI: [6]; World Bank ICP 2021 DataBank: [7]).

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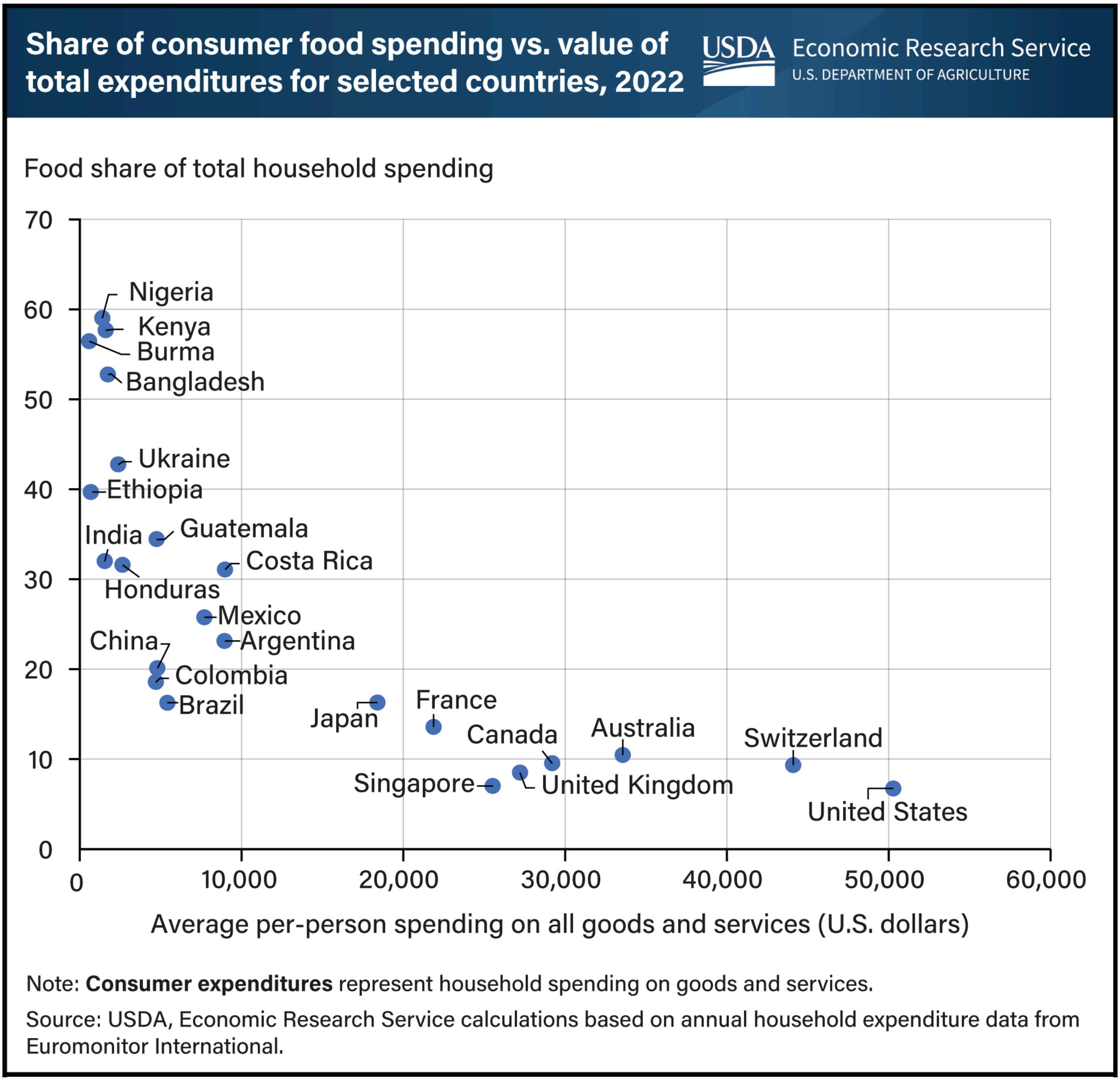
3) Cross-country evidence: levels and shares move in opposite directions

3.1 Absolute levels: higher overall expenditure → higher at-home food spending (nominal US$/person)

OWID's cross-country visualization "Annual food expenditure vs. total consumer expenditure" (USDA ERS-based) shows a **strong positive relationship** between **total consumer expenditure per person** and **annual at-home food expenditure per person** ([3]; framing and caveats: [1]). OWID also notes the x-axis is **log-scaled by default**, which changes how slope and dispersion appear (same chart link).

Benchmarks explicitly stated in OWID's text (not PPP- or inflation-adjusted):

- Below **$5,000** total expenditures/person/year: typically **< $1,000** at-home food/person/year.
- Above **$10,000** total expenditures/person/year: commonly **$2,000+** at-home food/person/year. (OWID: [1].)



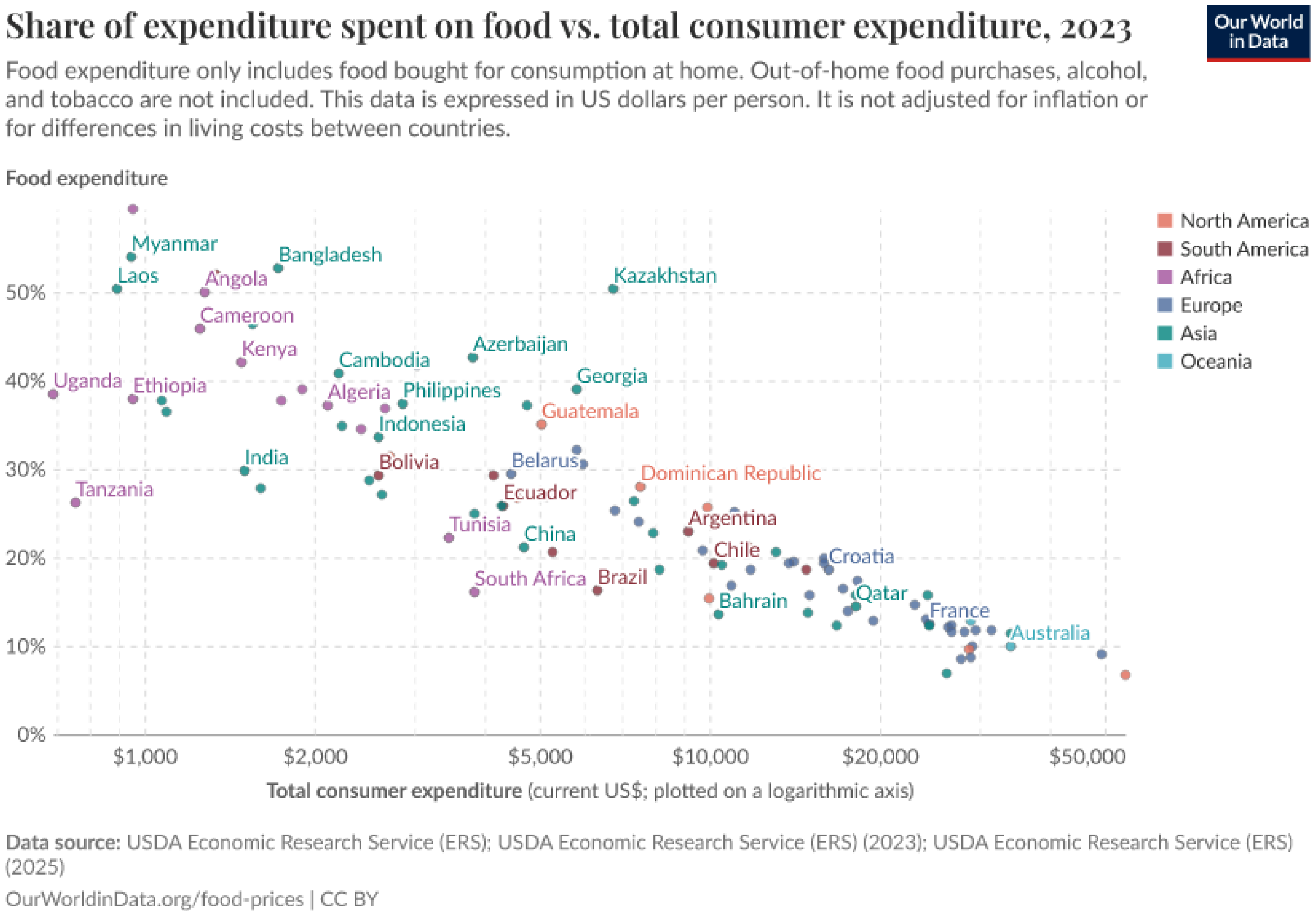
How to read this evidence. This chart supports only a *specific* claim: with OWID's definitions, richer/high-expenditure countries generally appear at higher **nominal at-home food spending** levels. Because values are in **current US$** and exclude out-of-home food, it does *not* by itself establish cross-country differences in *real* food quantities, total food spending, or affordability (OWID caveats: [1]).

3.2 Shares (Engel's law): higher overall expenditure → lower food *budget share*

OWID's companion chart "Share of expenditure spent on food vs. total consumer expenditure" visualizes the core Engel relationship in shares: as overall expenditure rises, the **food share declines** even though absolute food spending rises ([8]; OWID narrative and definitions: [1]). OWID notes the x-axis is on a **log scale** and that food here remains **at-home only**.

Rule-of-thumb share thresholds stated in OWID’ s text:

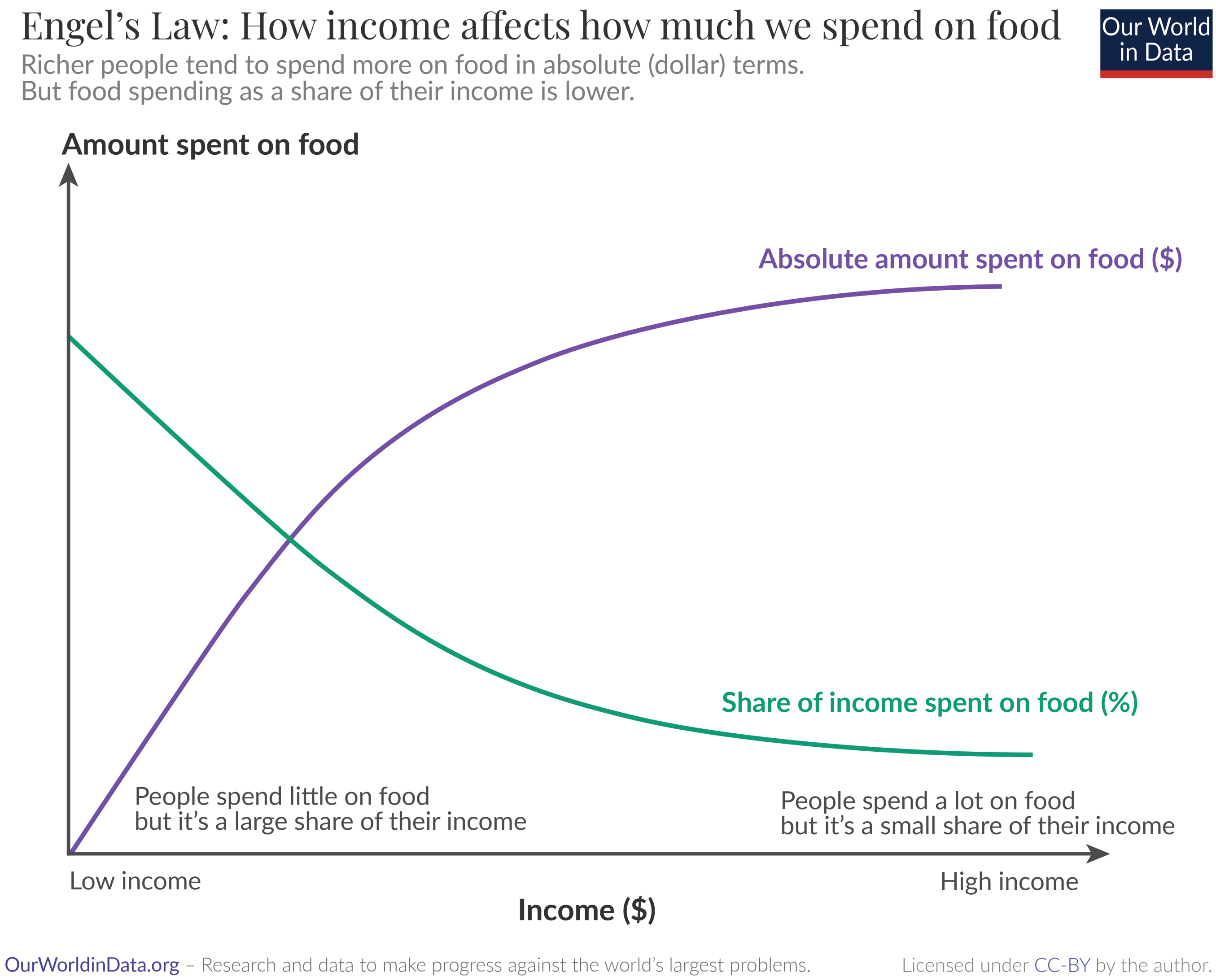
- Below **$10,000** total expenditures/person/year: often **≥25%** spent on at-home food.
- High expenditure levels: often **≤10%** spent on at-home food.
  (OWID: [1].)



Interpretation boundary. This is strong evidence for the *direction* of the Engel pattern across countries, but the **level** of any country’ s food share depends on (i) whether out-of-home food is excluded, and (ii) whether totals are measured in nominal dollars rather than in a price-level–adjusted framework (OWID caveats: [1]).

3.3 A single figure tying “levels up, shares down” together

OWID’ s Engel’ s law page also uses a combined illustration of the two relationships—absolute food spending rising and the food share falling with income/expenditure—to explain why the pattern is not contradictory (OWID: [1]).



3.4 External benchmark: USDA ERS "Chart of Note" (threshold-style food share examples, 2022)

USDA ERS provides a threshold-oriented cross-country summary emphasizing how food shares concentrate at high levels in poorer settings (ERS "Chart of Note": [9]).

From the ERS chart text as used in the source material:

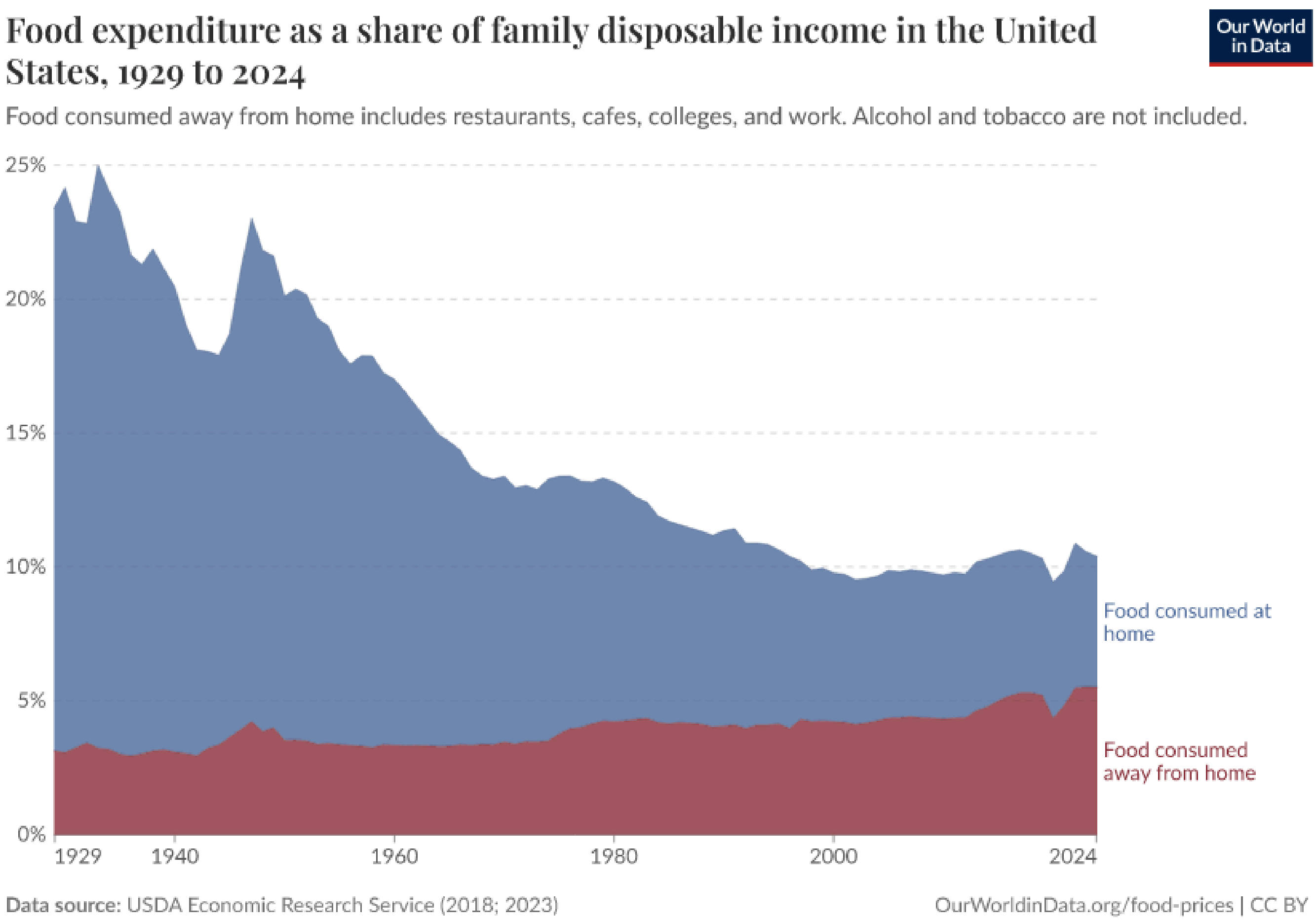
- **Low-income African and South Asian countries**: **>40%** of total consumer expenditures spent on food in **2022**.
- Some countries including **Nigeria, Kenya, Burma (Myanmar), and Bangladesh**: **>50%** in **2022**.
- Mid-range examples: **Costa Rica, Honduras, Guatemala** at **>30%**; **Argentina, Colombia, Mexico** around **~22.5%** in **2022**.
- Higher-income economies (examples listed by ERS include **U.S., Switzerland, Australia, Canada**) have smaller food shares, with ERS attributing this to larger disposable incomes and shifting expenditure composition with development/urbanization.
  (All from ERS: [9].)

Why this matters for verification. Compared with scatterplots, the ERS chart provides **explicit threshold statements** (e.g., “>40%”, “>50%”, “~22.5%”) that are easy to audit directly against the published chart notes.

4) Within-country and distributional evidence: Engel gradients within the same national context

4.1 A within-country visualization (OWID): food share vs family disposable income

OWID provides a chart relating food expenditure share to family disposable income (image export link: [10]). It presents the same qualitative structure expected under Engel’s law: **higher income is associated with a lower food share** (subject to OWID’s chart definition and the general caution that chart labels and definitions determine what “food” includes).



4.2 Within-country studies cited by OWID: South Africa (auditable) and China (lead only)

- **South Africa (district-level):** Mulamba (2022) analyzes "households' share of food expenditure and income across South African districts" using multilevel regression, providing within-country evidence consistent with Engel's law ([4]).
- **China:** The report materials include a pointer to "Engel's law in China: Some new evidence" (Chen, 2022), but no stable link was captured here. In this rewrite, the China item is treated as **an unverified lead** rather than as a fully citable result.

4.3 Income gradients within a high-income bloc: EU-27 quintiles (Eurostat HBS via Dudek figure)

A ResearchGate-hosted figure by Hanna Dudek is titled "Food and non-alcoholic beverages share expenditure by income quintile in EU-27 in 2005" and cites Eurostat's HBS tabulation "structure of consumption expenditure by income quintile (hbs_str_t223)" ([11]).

What is explicitly stated in the figure context:

- "The average household expenditure share for food and non-alcoholic beverages consumption was about **16.6% in 2005**." (Same Dudek figure page.)

A truncated caption fragment also appears ("Food represented more than **22% of total ...**"), but because the sentence is incomplete in the captured material, this report does **not** infer which group/category/denominator it refers to.

Verification gap (kept explicit). The quintile-by-quintile numerical values (Q1–Q5) are not extracted in the provided materials; verifying them requires either (i) a clean read of the downloadable figure or (ii) direct Eurostat retrieval of `hbs_str_t223` (figure download link: [12]).

4.4 Why "food share" is used as a welfare signal—and why definitions still dominate interpretation

Dudek (2011) frames food-expenditure share as an applied welfare/income indicator aligned with Engel's law and studies its dependence on total expenditure and household size ([13]).

At the same time, the comparability of "food share" depends on how total expenditure is constructed and what is included/excluded. This broader issue is part of the measurement discussion in Stiglitz–Sen–Fitoussi (2009) (Eurostat-hosted): [14], and in practical welfare measurement guidance such as Deaton & Zaidi (2002): [15].

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5) Mechanisms and behavioral channels (what the cited sources actually support)

5.1 Diet diversification as prosperity rises (OWID mechanism link)

OWID connects falling food shares to changing diet composition—moving away from "cheaper staple crops" (e.g., cereals and tubers) toward more diverse food categories—and supports this with a chart relating GDP per capita to the share of dietary energy from cereals/roots/tubers ([16]). This mechanism is presented in OWID's Engel's law narrative ([1]).

Boundary condition. This mechanism speaks to *composition shifts* and constraints on staple-heavy diets; it does not, by itself, quantify the contribution of composition change vs price levels vs total expenditures to observed "current US$" spending patterns.

5.2 Price responsiveness and distributional differences (New Zealand demand system example)

Ni Mhurchu et al. (2013) estimate **price elasticities (not income elasticities)** for 24 food categories using an Almost Ideal Demand System (AIDS), based on New Zealand household economic surveys (2007/08 and 2009/10) plus Food Price Index data (2007 and 2010) (PMC full text: [17]; DOI: [18]).

Quantitative results reported in the paper and summarized in the materials:

- Own-price elasticities for major commonly consumed food groups mostly range from **−0.44 to −1.78** (with two exceptions) (Ni Mhurchu et al., 2013: [17]).
- Cross-price elasticities are usually small; only **31%** have absolute value **> 0.10** (same source).
- Excluding an outlier ("energy drinks"), **9 of 23** food groups show significantly stronger own-price elasticities for the **lowest vs highest** income quintiles, with an average difference of **−0.30** (95% CI **−0.62 to 0.02**); **6** are significantly stronger among Māori, average difference **−0.26** (95% CI **−0.52 to 0.00**) (same source).

How this links back to Engel's law. Engel's law describes how budget shares shift with income; the New Zealand study shows that **price changes** (and price policies) can generate **distributionally uneven behavioral responses**, which is consistent with why "food burden" can vary sharply across income groups even within one country.

5.3 A noted extension in the original report (retained with caution): global meta-analysis claim

The original report cites an IFPRI global meta-analysis and a "Food Demand Meta-Elasticity (FDME) database" as supporting evidence on declining income responsiveness with development, including a category-specific note that "food away from home" can behave differently. These claims are retained only at the level stated in that report section and linked to the cited sources:

- IFPRI blog post (Jan 9, 2026): [19]
- FDME database DOI: [20]

- IFPRI Discussion Paper: [21]

Because the supporting details are not reproduced in the provided learnings beyond these links, this report treats the IFPRI content as **supporting context** rather than as a fully cross-checked empirical backbone for the main Engel claims, which remain anchored in OWID/USDA ERS and the cited within-country studies.

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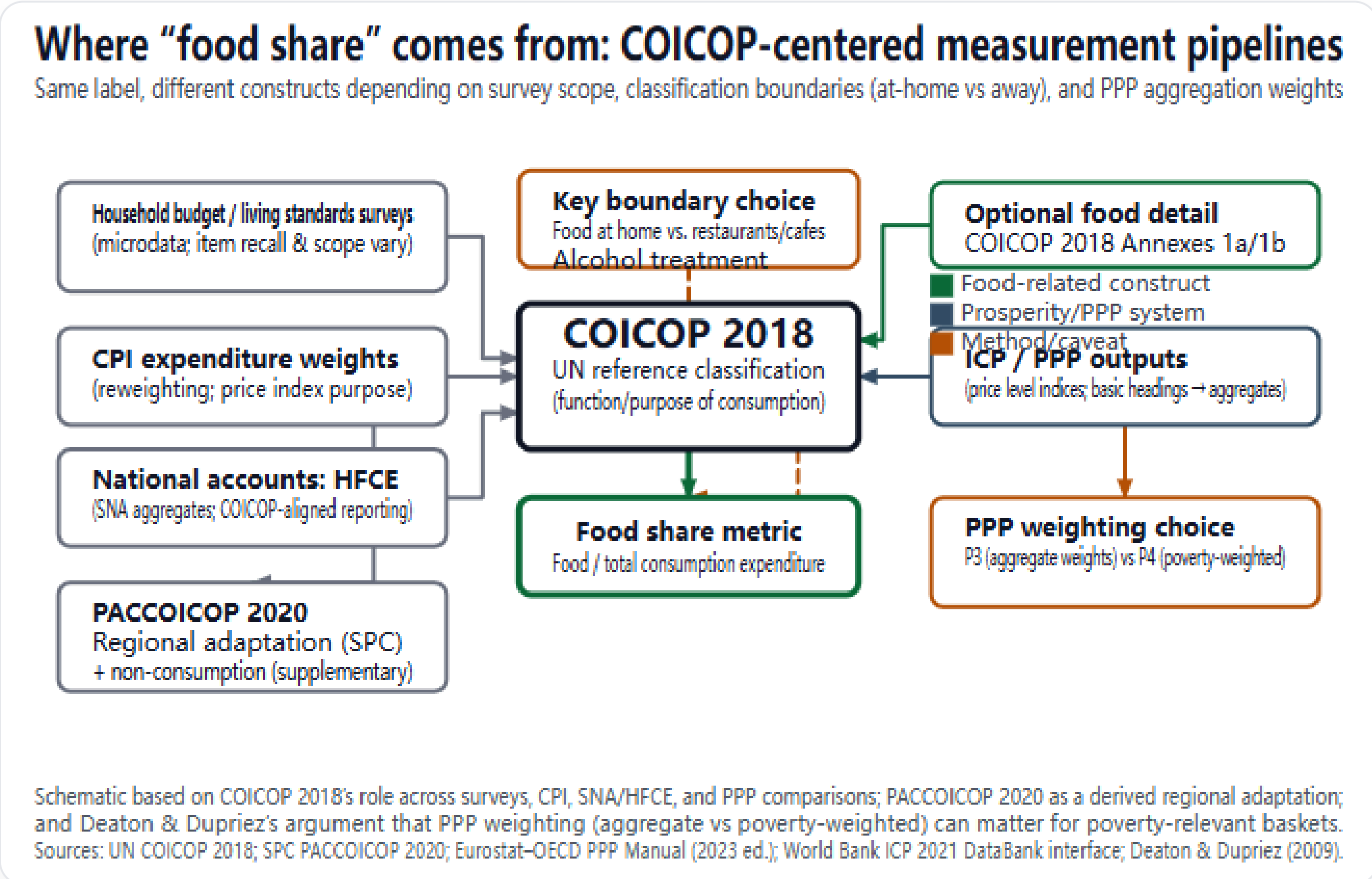
6) Measurement and comparability: why "food share" statistics are often not interchangeable

6.1 A practical "production pipeline" view: the same label can hide different constructs

"Food share" can be produced from household survey tabulations, CPI weights, national accounts (HFCE) aggregates, or PPP/ICP components mapped to consumption classifications. Treating these as interchangeable is a common source of error in cross-country comparisons.



6.2 Classification standard: COICOP 2018 as the reference for household consumption

The UN’ s COICOP 2018 is presented as the international reference classification for household consumption expenditure (UN COICOP 2018 page: [22]). The COICOP page also references optional high-detail annex structures for food products (Annex 1a/1b), which matters for reconciling fine-grained “food” constructs across surveys and statistical systems (same UN link).

Explicit gap preserved from the original report. This report does not quote the exact boundary-rule text separating “food” from restaurant/catering services because that specific excerpt is not captured in the assembled materials; therefore, it treats “at-home vs away-from-home” primarily as a **known comparability risk** rather than a quantified bias.

6.3 Regional adaptation example: PACCOICOP 2020 (Pacific) and boundary-rule uncertainty

PACCOICOP 2020 is described as derived from UN COICOP 2018, with the stated difference being inclusion of “non-consumption expenditure as supplementary information,” and is positioned for CPI weights, HFCE derivation, and poverty studies across Pacific Island Countries and Territories (SPC entry point: [23]; COICOP link: [22]).

The original report notes (and this rewrite retains) that the excerpted PACCOICOP materials do not provide enough operational detail here to confirm how “food away from home” boundaries are implemented in practice across countries—an important caveat when comparing food shares internationally.

6.4 Nominal vs PPP-adjusted comparisons: documentation exists, but series selection is still a step

For PPP methods and classification correspondence, the Eurostat–OECD methodological manual provides a documented reference point (DOI: [6]; licensing framework: [24]). The original report uses this mainly to support the claim that **classification correspondences and PPP computation methods are documented**, which matters when moving from nominal expenditures to PPP/price-level concepts.

For practical extraction, the World Bank ICP 2021 DataBank provides downloadable outputs (Excel/CSV/tabbed/metadata) and shows a multi-dimensional structure suitable for auditable workflows ([7], including the export/aggregation UI references cited in the original report: [25], [26]).

Open verification item retained. The original report notes it has not yet identified the exact ICP 2021 series code for a “Food and non-alcoholic beverages” price level index (PLI); this rewrite preserves that limitation and therefore does not introduce any PLI rankings or cross-country “real food price” claims.

6.5 Income-group comparisons: why group membership changes over time (World Bank classifications)

If analysts want to summarize food shares "by income group," the World Bank's country income classifications are updated annually (July 1) based on Atlas GNI per capita and inflation-adjusted thresholds (World Bank Data Blog: [27]; Atlas method explainer: [28]; SDR deflator explainer: [29]; indicator: [30]).

Implication for Engel summaries. A "low-income vs high-income" comparison is not only about different food shares; it also depends on which countries are in each group in the chosen year. The World Bank post documents that the distribution of reporting countries across income groups has shifted substantially between 1987 and 2023 (same World Bank blog link).

6.6 Even within one country: survey instrument differences (FoodAPS benchmark)

USDA ERS EIB-157 emphasizes that FoodAPS is a nationally representative "food acquisition + purchase" survey and documents that **FoodAPS estimates of total and food-at-home spending exceed CE estimates but are below NHANES estimates**, and that FoodAPS estimates a higher share of households with low/very low food security than other surveys ([5]).

Why this matters here. If "food spending" is survey-sensitive within the U.S., cross-country comparisons that mix pipelines (survey vs national accounts vs modeled series) must be explicit about source and method to remain interpretable.

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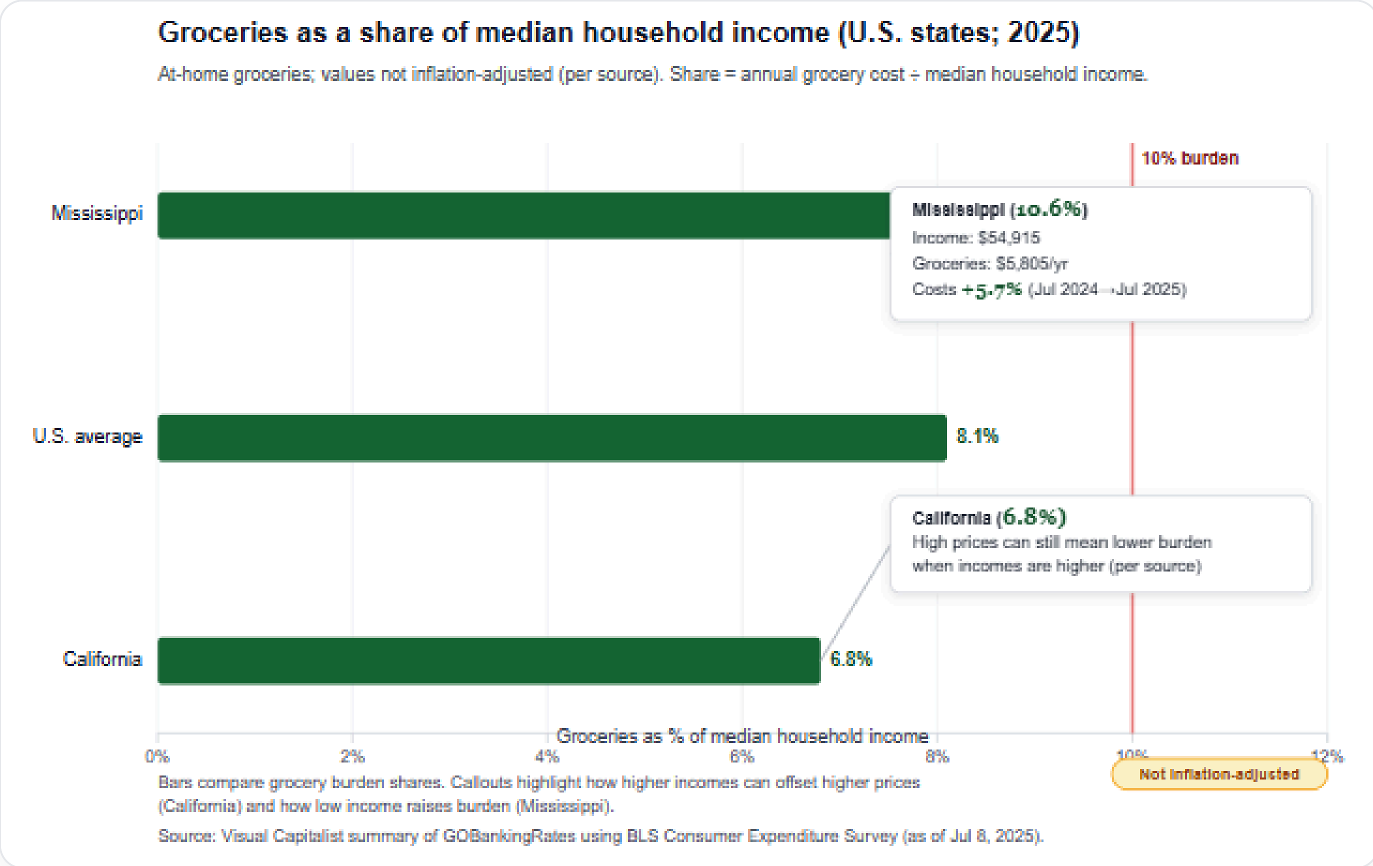
7) Communication extensions retained from the original report (burden framing and method sensitivity)

7.1 Share-of-income ("burden") framing: U.S. grocery costs by state

A state-level example illustrates why affordability discussions often track shares rather than levels. Visual Capitalist (summarizing GOBankingRates analysis using the BLS Consumer Expenditure Survey, as of July 8, 2025) reports:

- U.S. average groceries ≈ **8.1%** of median household income.
- California ≈ **6.8%**.
- Mississippi ≈ **10.6%**, with the arithmetic shown as **$5,805** annual grocery costs over **$54,915** median household income, and grocery costs up **5.7%** from July 2024 to July 2025.
  (Source: [31].)



7.2 Method sensitivity in global diet affordability metrics (retained as cited in the original report)

The original report also cites an affordability-method sensitivity example: Headey, Hirvonen & Alderman (2024) report that adding demographic scaling and an improved food-budget derivation reduces the estimated number of people unable to afford the EAT-Lancet reference diet from **3.02 billion** to **2.13 billion** (difference **0.89 billion**) (Food Policy DOI: [32]; PMC full text: [33]). The same paper notes additional sensitivity levers (age structure, price-data coverage/representativeness, physical-activity assumptions) (PMC link).

Connection back to Engel’s law reporting. This reinforces a general reporting rule already motivated by OWID’s caveats: figures should carry definition lines (what “food” includes; price basis; whether out-of-home is excluded) because headline counts and shares can move materially with method choices.

8) Open verification and research tasks (kept focused)

1. **Extract and verify EU-27 income-quintile values** for “food and non-alcoholic beverages” from Eurostat HBS `hbs_str_t223` (and reconcile with the Dudek figure): [11].

2. **Quantify the impact of excluding out-of-home food** in cross-country Engel comparisons by locating a comparable dataset including total food (home + away) or by mapping at-home vs restaurants boundaries under COICOP with explicit, citable boundary text (COICOP page entry point: [22]).
3. **Identify ICP 2021 "Food and non-alcoholic beverages" PLI series code** in the World Bank ICP DataBank before adding any PPP-based food price-level comparisons: [7].

References (sources cited in-text)

- Our World in Data (Hannah Ritchie, 2023), Engel's law topic page (archived): [2]
- Our World in Data, Engel's law page and linked charts: [1], [3], [8], [10], [16]
- USDA ERS, Chart of Note (2022 food share examples): [9]
- Mulamba, K. C. (2022), *Humanities and Social Sciences Communications*: [4]
- Dudek (figure; Eurostat HBS reference `hbs_str_t223`): [11]
- Dudek (2011), ResearchGate publication page: [13]
- Stiglitz–Sen–Fitoussi (2009), Eurostat-hosted: [14]
- Deaton & Zaidi (2002), World Bank document: [15]
- Pacific Community (SPC) entry point (PACCOICOP context as cited): [23]
- UN COICOP 2018 page: [22]
- Eurostat–OECD PPP Manual (DOI) and CC BY 4.0: [6], [24]
- World Bank ICP 2021 DataBank: [7]
- World Bank income classification (2024–2025) and methods: [27], [28], [29], [30]
- USDA ERS EIB-157 (FoodAPS benchmarking): [5]
- Ni Mhurchu et al. (2013), PLoS One (PMC; DOI): [17], [18]
- Visual Capitalist (U.S. grocery burden by state; citing GOBankingRates/BLS CES): [31]
- Headey, Hirvonen & Alderman (2024), *Food Policy* (DOI; PMC): [32], [33]
- IFPRI (as cited in the original report), FDME meta-analysis links: [19], [20], [21]

References

1. https://ourworldindata.org/engels-law-food-spending
2. https://archive.ourworldindata.org/20260325-171315/engels-law-food-spending.html
3. https://ourworldindata.org/grapher/food-expenditure-vs-gdp
4. https://www.nature.com/articles/s41599-022-01454-4
5. https://www.ers.usda.gov/publications/eib-economic-information-bulletin/eib157
6. https://doi.org/10.2785/384854
7. https://databank.worldbank.org/id/9c15e529
8. https://ourworldindata.org/grapher/food-expenditure-share-gdp
9. https://www.ers.usda.gov/data-products/charts-of-note/chart-detail?chartId=107494

10. https://ourworldindata.org/grapher/food-expenditure-share-family-disposable-income.png
11. https://www.researchgate.net/figure/Food-and-non-alcoholic-beverages-share-expenditure-by-income-quintile-in-EU-27-in-2005_fig1_259280953
12. https://www.researchgate.net/figure/Food-and-non-alcoholic-beverages-share-expenditure-by-income-quintile-in-EU-27-in-2005_fig1_259280953/download
13. https://www.researchgate.net/publication/259280953_Quantitative_analysis_of_the_household%27s_expenditure_for_food
14. https://ec.europa.eu/eurostat/web/products-manuals-and-guidelines/-/ks-78-09-717
15. https://documents1.worldbank.org/curated/en/099552210062217657/pdf/IDU0cf96acbc0a52d0d5ae0a9910d6d380c249c.pdf
16. https://ourworldindata.org/grapher/share-of-energy-from-cereals-roots-and-tubers-vs-gdp-per-capita
17. https://pmc.ncbi.nlm.nih.gov/articles/PMC3788811/
18. https://doi.org/10.1371/journal.pone.0075934
19. https://www.ifpri.org/blog/the-future-of-food-demand-evidence-from-a-global-meta-analysis-and-trend-projections/
20. https://doi.org/10.7910/DVN/7JRN8C
21. https://hdl.handle.net/10568/176595
22. https://unstats.un.org/unsd/classifications/Family/Detail/1220
23. https://www.spc.int/
24. https://creativecommons.org/licenses/by/4.0/
25. https://databank.worldbank.org/id/9c15e529#selectedDimension_DBList
26. https://databank.worldbank.org/id/9c15e529#aggregationRule
27. https://blogs.worldbank.org/en/opendata/world-bank-country-classifications-by-income-level-for-2024-2025
28. https://datahelpdesk.worldbank.org/knowledgebase/articles/77933-what-is-the-world-bank-atlas-method
29. https://datahelpdesk.worldbank.org/knowledgebase/articles/378829-what-is-the-sdr-deflator
30. https://data.worldbank.org/indicator/NY.GNP.PCAP.CD
31. https://www.visualcapitalist.com/grocery-costs-as-a-share-of-income-by-u-s-state/
32. https://doi.org/10.1016/j.foodpol.2024.102654
33. https://pmc.ncbi.nlm.nih.gov/articles/PMC11287492/